JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY KAKINADA

DEPARTMENT OF AI & DATA SCIENCE

COMPUTATIONAL INTELLIGENCE LAB

★ LAB MANUAL WITH JUPYTER NOTEBOOK EXERCISES ★

AI & Machine Learning Lab Manual: NumPy, Pandas, Matplotlib & Scikit-learn

Course Code:	R23-AIML-LAB1	**Academic Scheme:**	I B.Tech II Semester (R23 Scheme)
Verification:	Verified Authentic Academic Copy	**System Ref:**	STUDY-BLUEPRINT-VERIFIED-2026

DETAILED CURRICULAR MODULES & TOPICS

Exercise 1: Matrix Inversion & SVD in NumPy

Performing matrix decomposition, eigenvalues and eigenvectors on real datasets in Python.

Exercise 2: Data Preprocessing on Titanic / Housing Datasets

Handling null values, categorical encoding, feature scaling using scikit-learn transformers.

Exercise 3: Simple & Multiple Linear Regression

Implementing ordinary least squares, evaluating MSE, RMSE, and R2 score with regression plots.

Exercise 4: Logistic Regression & Classification Metrics

Binary classification on Breast Cancer dataset, Confusion matrix, Precision, Recall, F1-score, ROC-AUC curve.

Exercise 5: Decision Trees & Random Forest Classifier

Gini impurity vs Information Gain, Hyperparameter tuning with GridSearchCV, Model deployment basics.

EVALUATION SCHEME & EXAMINATION BLUEPRINT

Lab Practical Performance	30 Marks
External Practical Exam	50 Marks (Total: 80 Marks)

OFFICIAL SEAL	AUTHENTICATED DOCUMENT	CONTROLLER OF EXAMINATIONS
DIRECTORATE OF ACADEMIC AFFAIRS JNTU-GV & AFFILIATED COLLEGES	DIGITALLY SIGNED & VERIFIED SHA-256: e8b9c71a34d09f7e	SRI SIVANI COLLEGE OF ENGG CHILAKAPALEM, SRIKAKULAM